

MILAN

An owner, a clock and a receipt for every broken community asset

Demo pack · three parts

A · Problem statement — the night school that closed when the solar lights broke

B · Solution — title, abstraction, and the instance walked through the platform

C · Industry upload spec — what a funder files, and what gets returned to them

Field	Value
Parent problem statement	SIH26043 — A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships
Sponsoring organisation	Government of Jharkhand · Dept. of Higher & Technical Education
Theme · Category	Smart Education · Software
Portal deadline (idea submission)	30 September 2026
Demo instance	Jharia block (DHN-JHA), Dhanbad district — night classes for working children
Domain · severity · priority	EDUCATION · 0.82 (S2) · 66.43 / 100 (S4, zero model calls)
Routed to	NIT Jamshedpur · BIT Mesra · BIT Sindri (claimed) — top 3 distinct orgs
Seed data used	12 HEIs · 51 capability labs · 24 districts · 263 blocks · 8 industry orgs
Team ID / institution	(fill in before upload)
Document built from	docs/demo/*.md @ 608cf54

How to read this pack. Parts A and B are the submission text: the problem as a citizen files it, and the platform answer to SIH26043, with the arithmetic shown rather than asserted. Part C is the operating rule for the industry side — which is where challenge platforms quietly die, because a funder with no evidence obligation produces a press release instead of an outcome. Figures marked *[demo]* are seeded for the demonstration; the geography, institutions, companies and laboratory capabilities are the rows this repository actually runs on.

1. Communities hold the problems; universities hold the researchers. We are the pipeline between them.
2. CPGRAMS routes complaints to officers. Milan routes unsolved problems to labs, with a clock.
3. Discovery is never luck: every problem is pushed to matched departments, and every state escalates.
4. 200,000 Indian students invent a fake final-year project a year. We give them real ones.
5. We do not stop people from sharing work. We make it impossible to erase who did it.

PART A — PROBLEM STATEMENT

Problem Statement — "The lights went out and the night school closed"

Demo instance of SIH26043 · A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships **Government of Jharkhand** · **Theme: Smart Education** · **Category: Software**

Status of the numbers in this document. Everything marked [demo] is a seeded figure invented for the demonstration and must be replaced with field data (Census child-work tables, NLSS, UDISE+, the district LED/SSA asset register, the Panchayat's works records) before this is filed anywhere real. The geography, institutions, companies and laboratory capabilities are **not** invented — they are the rows this repository actually seeds, and the demo runs against them.

1. Title

Night classes for working children stopped when the solar lights broke — and nobody was accountable for repairing them.

Working title for the platform record: Solar lighting and the missing owner: restoring night classes for working children in Jharia block.

2. Identifiers

Field	Value
Parent problem statement	SIH26043 (Software · Smart Education)
Sponsoring organisation	Government of Jharkhand, Department of Higher & Technical Education
Instance location	Jharia block (DHN-JHA), Dhanbad district, Jharkhand
Domain	EDUCATION (secondary: ENERGY, ACCESSIBILITY)
Recurrence	constant
People-affected bucket	100–1,000 → stored as midpoint 550
Block vulnerability index (seeded)	0.87 — the highest of the 11 seeded Dhanbad blocks

3. Background

In the Jharia coalfield, school is not the end of a child's day — it is the second shift. Children who sort coal at the washery chowk, load trucks at dusk, mind goats on the subsidence edge, or work in tea stalls and domestic service in Dhanbad's wards attend **evening bridge courses** run by volunteer teachers, an NGO-run night school, and an open-schooling support centre. Four centres, roughly 500 children between them [demo], meet from 18:00 to 20:30.

The only light in those rooms, and on the lanes that lead to them, is **solar**. Street lights installed under a state/central rural-lighting scheme, plus small rooftop solar kits with battery boxes bolted under the veranda of two of the four buildings.

Fourteen months ago the lights began failing [demo]. First one centre, then the lane, then all four. The causes were ordinary and identical across the state: charge controllers burned out in the last pre-monsoon, batteries swelled past end-of-life, two battery boxes were stolen, one pole was hit by a truck, and no one held the contract any more — the installation agency's one-year defect liability period had expired in month eleven.

What happened next is the actual problem. Nobody owns a solar street light. The **education** department owns the children but not the lamps; the **Panchayat / ULB** owns the lamps but budgets nothing for their repair and holds no asset register that says which pole is where; the **scheme portal** that funded them records installation, not maintenance; the **electrical contractor** who installed them has moved to another block. When the children's attendance collapsed, each of those four bodies was correct: not my asset, not my budget, not my department.

By the third month, three of the four centres stopped running after dark entirely [demo]. In an informal coalfield economy, a child with an unlit evening is a child at the chowk. Attendance at the remaining centre fell and did not recover; the two centres that had enrolled girls — 60% of their roll — closed first, because the lane is the barrier, not the classroom.

4. Statement of the problem

The broken lamp is not the problem. The missing owner is.

Jharkhand has thousands of community assets — lights, pumps, water taps, angular-road culverts, solar kits on school roofs, toilets, anganwadi roofs — that sit in exactly this gap: installed by a scheme, handed to a body with no repair budget, monitored by nobody, and invisible to every department whose work they gate. Where a citizen's report has nowhere to go, the asset stays broken and the outcome the asset existed to produce — children in class after dark — quietly dies. No dashboard shows the death, because no system links the asset to the outcome.

The problem this instance puts to the platform (SIH26043) is therefore **a routing and accountability problem in a Smart Education setting**:

How does a platform take a vernacular citizen report — "the night school closed because the lights broke and no one repairs them" — categorise it, judge whether it is a grievance (a repair someone already owes) or an unsolved problem (no known fix, no owner), route it to a university team capable of producing an answer, put a clock on the whole thing so it cannot die silently, and let industry fund the deployment with evidence an auditor can check — while proving every credit and every number?

5. What makes it hard (and why it is a research question, not a tender)

1. **Grievance-or-problem ambiguity.** If one light on a Panchayat road is out, that is a complaint with a known fix and a defined office. This is four centres dark for fourteen months with no live contract — there is no officer to complain to, because the accountable thing has expired. The platform must separate the two and forward the first while routing the second. (Milan: S1 triage + FORWARDED_EXTERNAL.)
2. **The fix is not the lamp.** Replacing 40 lights costs money the block already spent once. The defensible answer is a reparability model: which failure modes are field-fixable by a trained local electrician, what a ₹-scaled spares corpus looks like, whether a ₹300-500 retrofit controller with 1 bit of telemetry ("this pole drew current last night") is worth the data plan, and what the Panchayat will actually sign up to run. That is a legitimate student research project with a measurable dependent variable: **class-nights held after dark**.
3. **Multi-department.** Civil works, solar, school education, ICDS, rural development, the district mineral foundation, and CSR money all touch it. No single department can be routed to.
4. **Equity visibility.** 500 children across four centres is small in absolute terms. A scoring function that rewards headcount linearly will never surface it; the one that does must be log-scaled and must weight chronic recurrence and block vulnerability. This instance is the proof case for that design choice.
5. **Darkness is the dropout variable, and it is not in anyone's data.** Nobody records lane lighting and night-school attendance in the same table. Establishing the link is part of the research.

6. Requirements placed on the platform by this instance

#	Requirement (from SIH26043)	What this instance specifically demands
R1	Intake for citizens/local bodies with photos, location, documents	A 24-year-old washery worker must be able to file it in Hindi from a ₹8,000 phone, with two photos of a dead pole, no login required to be believed, and a tracking ID she can say over the phone
R2	AI categorisation by thematic domain	Must land on EDUCATION and not PUBLIC_SERVICE/URBAN_INFRA, with severity above the human-gate threshold and a low capital-works flag — or it never reaches a lab
R3	Deduplication as signal	Five neighbours + one teacher reporting the same four dead centres must merge into corroboration, not five competing tickets
R4	Routing by institutional capability	Must find electrical/solar + embedded-systems + education-department labs within reach, not the nearest college with a CS department
R5	University workflow	A multidisciplinary team (EEE + ECE + Education), faculty mentor, milestones, named students
R6	Industry collaboration	Must attract a mining/utility CSR that funds hardware, and hold it to evidence
R7	Lifecycle + SLA	14 months of silence is the failure being fixed: every state gets a clock and an escalation
R8	Analytics for government	Domain distribution must show that EDUCATION items are being lost, and the district board must see this one at the top while it is unclaimed
R9	Verification of outcomes	A funder's "lights restored" must not count until the teacher and the children's families say class is happening again
R10	Offline & vernacular	Jharia has dead zones; the pipeline must degrade to deterministic rules rather than fail

7. Expected solution components (mirrors SIH26043, applied)

- Citizen engagement module** — multilingual intake with photo evidence, device-side privacy blur, GPS or place search, a bucket for scale rather than a fake precise count, and a self-review step where the reporter approves the platform's re-framing or rejects it.
- AI-enabled problem management** — triage (safety / grievance), classification (11 domains, severity, solvability, capital-works flag), dedup+corroboration via embeddings, deterministic priority scoring with the arithmetic on screen, capability routing with a written reason per institution.
- University collaboration module** — offers pushed to three matched labs, claim window, multidisciplinary team formation by name, mentor assignment, milestones, publication of the artifact with a content hash.
- Industry partnership module** — expression of interest against a live challenge, acceptance by the project lead, funded-pilot record, and the implementation claim (see Part C for what industry uploads).
- Project lifecycle management** — one state machine owning every transition; SLA ladder; dispute handling; closure only after verification.
- Visual analytics** — district board, domain funnel, confirmation gap, public bounty board, public ledger, \$135 CSR export.
- Notifications** — push, never browse: each notice points at the one thing to act on.

8. Outcomes this instance should be judged on

Measure	Baseline [demo]	Target
Class-nights held after dark, per month, across the 4 centres	11	≥ 90
Reported-broken → confirmed-repaired median days	not measured; 14 months on record	≤ 21 days
Assets with a named owner + repair contact in a public register	0 of 46	46 of 46
Night-school enrolment (girls)	40% of prior roll	restored within one term
Centres with any functional lighting telemetry	0	≥ 2 (retrofit, read-only)
Citizen-confirmed impact (not funder-claimed)	0	the only number that counts

9. Stakeholders

Primary: the ~500 children [demo] and their families; the 6 volunteer teachers; the night-school committee. **Institutional:** Dhanbad district administration; Jharia block/Panchayat samiti; the school education department's dropout-cell; SCERT (open-schooling support). **Technical:** the coalfield companies and utility CSR arms operating in-district; BIT Sindri, NIT Jamshedpur, BIT Mesra, IIT (ISM) Dhanbad labs; the state renewable-energy development agency. **Adjacent:** the electrical works contractor community; Self-Help Groups that could run a tariffed repair corpus.

10. Data, interfaces and constraints

- **Available at filing:** photographs with device GPS (to be stripped on upload), the Panchayat's works register if it can be photographed, scheme-era installation records (paper), NGO attendance registers (paper), no digital asset register for the lamps.
- **Constraints:** intermittent connectivity; Hindi-dominant (Santali and Bangla present); no assumption of a smartphone per household; no government write-API available for asset registers; a citizen cannot be asked to know which department owns a pole.
- **Privacy hard line:** the beneficiaries are minors, several of them in child labour situations. No face imagery, no named child lists, no home locations. Blurring happens on the reporter's device and the unblurred original never leaves the phone.
- **Evaluation criteria a judge should apply to any submission:** correct grievance-vs-problem routing; explainable priority (not a black-box score); a clock on every state; a credit ledger nobody can erase; an outcome that only the beneficiary can confirm; and a working offline path.

PART B — SOLUTION

Solution — MILAN: Koi Maalik (an owner, a clock, a receipt for every broken community asset)

Companion to: docs/demo/PROBLEM_STATEMENT.md (Jharia night classes, Dhanbad) · **Upload spec for the industry side:** docs/demo/INDUSTRY_UPLOAD_SPEC.md

1. Solution title

Primary (recommended):

MILAN — Koi Maalik: an AI-routed assignment engine that turns citizen problems into funded university research

Alternates for different field limits:

Title	Chars	Use when
MILAN — Koi Maalik: an AI-routed assignment engine that turns citizen problems into funded university research	110	unconstrained field, or your own college's form
MILAN — Koi Maalik: routing citizen problems to labs, with a clock and a credit ledger	86	general submission
MILAN: an owner, a clock and a receipt for every broken community asset	71	safe default — fits an 80-char limit
Koi Maalik — restoring the night school by giving the lamp an owner	67	if you want the story in the title

Counts are measured, not estimated, because the SIH idea form truncates silently and a title cut mid-word in the reviewers' list is a needless loss. If you are unsure of the limit, use the 71-character row.

"Koi maalik" is the Hindi a judge in this room will already be thinking: koi maalik nahi tha — nobody owned it. Naming the intervention after the absence is the whole pitch in two words.

2. Abstraction

2.1 Portal abstract — 150 words exactly (measured; meets a 150-word cap)

Five hundred working children in Jharia block stopped attending night school when the solar lights broke. Nobody repaired them in fourteen months — not for want of parts, but because no department owns a lamp. MILAN closes that gap. A citizen files the problem in Hindi with two photographs; an AI pipeline triages it, separates a grievance someone already owes from an unsolved problem, classifies it across eleven domains, merges duplicates as corroboration, and scores priority with six deterministic weighted terms, printed on the public page. It is then pushed to the three departments best matched on capability, distance, capacity and track record, each with a written reason the model may not invent. Every state carries an SLA clock that escalates in public, every contribution is sealed in an append-only hash-chained ledger. Universities research and publish; industry funds the pilot with auditor-grade evidence. Impact counts only when the community confirms it.

2.2 Extended abstract

The insight. Every stakeholder in this story is behaving correctly. The education department funds teachers, not lamps. The Panchayat inherited 46 poles with no repair line item. The installing

contractor's defect-liability period expired in month eleven. The scheme portal reports lights installed, which is true. Each organisation is right, and the school is closed — which is the definition of an institutional gap rather than an engineering one. No repair backlog in India lacks for spare controllers; it lacks for an owner, a deadline, and a record that survives the person who cared.

What MILAN does. It is a **routing and accountability layer** placed over the gap, not another asset registry, and not a grievance portal. CPGRAMS and JharSewa route a complaint with a known fix to an accountable officer. There is no officer here — the fix is unknown and the ownership has expired, so the report has nowhere to go. MILAN routes unsolved problems to laboratories, with a time-bound assignment, and forwards the ones that were actually grievances to the right portal, telling the citizen exactly where the file went.

How it works end to end for this instance. A washery worker files it: Hindi, two photos, a pin on the lane. The six-stage pipeline runs in about eight seconds (P0 language → S1 safety/grievance triage || S2 domain/severity → S3 dedup → S4 priority → S5 routing). Six neighbours and a teacher report the same four dark centres; S3 merges them into corroboration rather than six competing tickets and credits both sides. S4 scores it **66/100**, above the routing bar, and the breakdown — severity, children affected, Jharia's vulnerability index, corroborations, chronic recurrence, no endorsement yet — is visible to everyone with no login. Severity 0.82 is over the 0.70 threshold, so nothing is sent until the District Collector releases it at the human gate. It routes to **NIT Jamshedpur's Power Systems Lab**, **BIT Mesra's Power and Energy Systems Lab**, and **BIT Sindri's Embedded Systems & IoT Lab** — chosen on capability, distance and declared capacity, not on name — with BIT Sindri claiming within the 7-day window. The deliverable is not "reinstall 40 lights." It is a field-fixable failure taxonomy, a sub-₹500 telemetry retrofit that lets a Panchayat see which poles drew current last night, a costed repair corpus and a local electrician protocol the Panchayat will actually sign, and a policy brief on writing O&M into the scheme at sanction time — published as a hashed artifact that becomes prior art nobody can patent over. **Damodar Valley Corporation** sees a live, citizen-endorsed need inside its own district focus, funds the retrofit through the industry module, and files an evidence pack (Part C). It marks the work implemented. That is a **claim, not an outcome**: the counter moves only when the teacher and the children's families answer yes, class is running again on a signed link that needs no login.

Why it is fair. Jharia's 500 children lose to any town's 6,000 under linear scoring, so people-affected is log-normalised and block vulnerability carries 18% of the weight — the two design choices that make a small, chronically broken thing in a high-vulnerability block reach a lab at all. A duplicate report is signal, never noise. An unclaimed problem escalates in public instead of expiring quietly. And a student who credited a night-school pilot cannot have their name edited out by whoever funds it next.

Why it is deployable. One Next.js 15 deployable on Vercel (Mumbai), one Postgres 17 with pgvector doing the work of rows, vectors, full-text search and the event outbox, three AI providers falling to a deterministic rules tier that makes **zero network calls** — demonstrated offline at fallback level 2 for every one of 53 model calls. No blockchain, no Kafka, no separate vector database, no fine-tuned model: a SHA-256 hash chain plus a public timestamp gives the same non-repudiation for free, and we say so on the slide rather than hiding it.

3. The instance, walked through the platform

Step	Where it happens	What is true afterwards
1 · File	/submit (or POST /api/intake)	Row created at SUBMITTED; photos blurred on-device, EXIF stripped, objects keyed by content hash; tracking ID sayable on the phone; credit edge + PROBLEM_TEXT ledger entry + SLA deadline written in the same transaction
2 · Language	P0	body_en working copy added beside body_original, never replacing it

Step	Where it happens	What is true afterwards
3 · Triage	S1 → lib/ai/triage.ts	Verdict CONTINUE. Not a grievance: there is no live work order and no sanctioned repair, so it is not forwarded to CPGRAMS. Had the report named a sanctioned scheme with a pending contractor, it would have been FORWARDED_EXTERNAL with the exact payload rendered publicly
4 · Classify	S2	Domain EDUCATION, severity 0.82 , solvability high, capital-works flag false → a research question, not a tender
5 · Embed + dedup	S3	cosine ≥ 0.86 auto-merge, 0.72–0.86 one adjudication call; 6 reports merge into the oldest survivor; corroboration_count = 6; both sides credited; nothing discarded
6 · Score	S4 (packages/scoring)	66.43/100 , zero model calls, six terms below, SCORING_VERSION 2.0.0 stamped
7 · Route	S5 → gate	Shortlist computed; severity ≥ 0.70 → offers written with notified_at = null; /gov/gate releases → ROUTED, 7-day claim clock
8 · Claim	/hei/.../claim	projects row, 4 students credited by name , mentor edge, and the reporter on the team as Domain Informant; other two offers expire; capacity decrements
9 · Research	/hei/projects/[id]	PROPOSAL_APPROVED → IN_RESEARCH; every write resets last_activity_at, which is what the silence ladder measures
10 · Publish	artifact panel	File keyed by its own SHA-256, REPORT ledger entry; title and abstract public regardless of licence ; RESTRICTED files served only against a logged, purpose-stated request
11 · Fund	/industry/...	EOI → acceptance → FUNDER credit edge → MoU generated, hashed into the ledger, explicitly an unsigned draft
12 · Deploy	markImplemented()	IMPLEMENTED = a claim. Renders grey as claimed, not confirmed everywhere, including the \$135 export
13 · Verify	/me/verify/[token]	YES → CITIZEN_VERIFIED, the only event that moves the impact counter; PARTLY → its own bucket; NO → DISPUTED, Collector notified, counter unmoved
14 · Close	7 days later	CLOSED by the reaper — bookkeeping is on a clock too

3.1 The priority score, as the public page shows it

```

SEVERITY raw 0.82 (S2) ×0.28 = 22.96
PEOPLE AFFECTED raw 550 (bucket 100–1,000) → log(1+550)/log(1+100000) = 0.5482 ×0.20 = 10.96
BLOCK VULNER. raw 0.87 (Jharia, seeded) ×0.18 = 15.66
CORROBORATIONS raw 6, mean trust 0.50 → ×1.00 → √6/√50 = 0.3464 ×0.14 = 4.85
RECURRENCE raw "constant" = 1.00 ×0.12 = 12.00
ENDORSEMENT raw 0 (no block officer yet) ×0.08 = 0.00
TOTAL = 66.43 / 100

```

Add one field endorsement at /gov/verification and the number becomes **74.43** — eight points, visible, computed by hand on the page. That is the answer to "how do we know the score isn't arbitrary?": it is not objective, it is six declared weights on a versioned, unit-tested pure function, and a state authority can change them without a redeploy.

3.2 Why these three labs (S5, indicative)

#	Lab	Sem	Ovl	Dist	Cap	Hist	Reason sentence the guardrail allowed
1	NIT Jamshedpur · Power Systems Lab (energy solar micro-grid rural-electrification load-monitoring)	0.79	0.62	0.64 (110 km)	0.40	0.55	"Strongest fit on solar and rural electrification, 110 km away, 2 slots open."

#	Lab	Sem	Ovl	Dist	Cap	Hist	Reason sentence the guardrail allowed
2	BIT Mesra · Power and Energy Systems Lab (renewable power-electronics)	0.76	0.55	0.66 (105 km)	0.40	0.45	"Matches solar and power-electronics specialisation, 105 km away, 2 slots."
3	BIT Sindri · Embedded Systems & IoT Lab (iot low-cost-sensors telemetry loran)	0.68	0.48	0.95 (12 km)	0.40	0.35	"Closest at 12 km, with low-cost telemetry capability for the retrofit."
—	near-miss: Ranchi University · Dept of Education (dropout pedagogy)	0.41	0.71	0.63	0.40	0.30	ranked 4th: highest pedagogy overlap, insufficient electrical capability

Weights: semantic 0.45 · specialisation overlap 0.20 · distance 0.15 ($\exp(-\text{km}/250)$) · declared capacity 0.12 · track record 0.08 (Laplace-smoothed, prior 3). Top **3 distinct organisations**, never two labs from one college. The model writes only the sentence, and only around the top three contributing terms; guardReason() rejects any number in that sentence that is not in the facts it was handed, falling back to a template. The near-miss row is the demo's best slide: the platform can explain a fourth place as honestly as a first.

4. What the university actually delivers (so the assignment is gradeable)

1. **Failure taxonomy + field diagnostic protocol** for the three lighting architectures present in the block, derived from teardown of the failed units — the deliverable a Panchayat electrician can use.
2. **A $\leq ₹500$ retrofit controller** (current-draw or Hall-sense + LoRa/GSM one-bit heartbeat) proving that "did pole 17 light last night?" is answerable without a human walking 46 poles; bill of materials, firmware, and the failure modes it cannot detect, stated.
3. **A costed O&M model:** spares corpus vs AMC vs SHG-run repair crew, at block scale, with the number that makes the difference — the ₹-per-pole-per-year that a Panchayat has previously never been asked to budget.
4. **The evidence that darkness, not disinterest, caused the dropout:** attendance registers against a lighting timeline, with the girls-only effect isolated.
5. **A policy brief:** O&M corpus and asset-ledger fields written into the sanction, not the handover.
6. **Published artifact**, hashed into the ledger, licence CC_BY (recommended) or RESTRICTED — and under either, the existence of the work stays public.

5. Design commitments this instance exercises

Invariant	What it does here
1 · nothing dies silently	22 non-terminal states all carry a deadline row; a CI query fails the build if one is missing. 14 months of nobody owns it is precisely the failure this CI check exists for
3 · AI proposes, code decides	0.82 severity is a model output; the 66.43 that routes it is arithmetic in packages/scoring
5 · human gate	the collector, not the model, decides that a child-labour-adjacent education claim goes public
7 · counter moves at CITIZEN_VERIFIED	DVC's "lights restored" is grey until the community says class is running
9 · duplicates are signal	six reports → one survivor with six corroborations and six credits
10 · every number clickable	the table in §3.1 is on the public challenge page, no login
8 · offline-safe	village-level connectivity in the coalfield is not assumed anywhere on the demo path

6. Novelty — and the honest comparison

What exists	What it does	Why this is different
CPGRAMS / JharSewa	complaint → accountable officer with a defined remedy	here there is no accountable officer and no defined remedy; MILAN detects that case and forwards the ones where it is a grievance
Asset portals / smart-city dashboards	record installation and outages of assets	they measure the asset , not the outcome the asset was bought to produce (class-nights held)
SIH/ISFC-style innovation events	episodic challenges, prize at the end	the challenge has a lifecycle, a clock, and a verification loop; the assignment outlives the demo
CSR project reports	self-reported beneficiaries and spend	three separate counters — confirmed / partly / claimed-not-confirmed — never summed into one flattering number
A university's own consultancy cell	institution-initiated, supply-driven	demand-driven: the community's problem, scored and pushed, with the citizen on the team

7. Feasibility, scale and cost

- **Build status:** this is a working application, not a plan — 4 roles + admin, 5 seeded accounts, a six-stage pipeline with per-call receipts in `ai_runs`, an append-only ledger enforced by a database trigger, an SLA reaper with four ladders, a \$135 export, and a judge console at `/demo` that drives the lifecycle with the production server actions.
- **Marginal cost per challenge:** ~53 model calls across five stages, all flash-tier or cheaper; embeddings cached on input hash so a replay is free; the rules tier costs nothing. Storage is content-addressed, so a duplicate upload is not a second object.
- **Scale:** one deployable; `pgvector` HNSW handles millions of rows; modules are folder-local so pipeline, ledger, SLA and notifications split into services without a rewrite. Multi-state is a data load (districts/blocks/orgs), not a migration.
- **This instance at state scale:** 263 blocks × ~20–50 community assets each is the addressable ledger; the platform only ever needs the reports, not the inventory, to start.

8. Declared limits (say them before someone finds them)

No native app or offline PWA sync (responsive web at 320px + `localStorage` drafts) · no video attachments (≤ 3 photographs) · SMS/WhatsApp delivery and IVR intake are seams with mock inboxes, not gateways · Hindi and Santali copy unverified by a native speaker for this demo · attendance and asset figures in this document are seeded, not surveyed · Jharia block resolution is nearest-centroid inside a polygon-resolved district, and no PMTiles basemap is loaded, so the map says so on screen · e-signature and payment rails are out; the MoU is generated, hashed and labelled an unsigned draft on its own face.

9. Next, if this is funded

v1.1 — asset-class ontology so "which pole" joins to a public registry; a repairability score as a seventh scoring term; a native-language ASR path in production (the stage exists, the demo uses a seeded transcript); self-serve institution and CSR onboarding. **v2** — block-level boundary geometry, real CPGRAMS write API, a third-party timestamp on the ledger anchor, and the O&M-corpus instrument itself with the Panchayat as counterparty — which is the point: the research output becomes a budget line, or it was a poster.

PART C — INDUSTRY UPLOAD SPEC

Part C — What an industry partner uploads, and why

For: the CSR / sustainability team of a company responding to a MILAN challenge (worked example: Damodar Valley Corporation funding the Jharia night-school lighting retrofit, or BCCL / SAIL / JSPL / Tata Steel Foundation on any in-district challenge).

The principle in one line: the platform accepts **evidence, not marketing**. Every file a funder uploads either (a) commits money, (b) describes what was installed, (c) proves it works, or (d) names the person who will keep it working. Anything else is a press release, and it belongs nowhere near a §135 report.

0. Before you can upload anything (gate 0)

Requirement	Why
An INDUSTRY account whose organisation is admin-approved (orgVerificationStatus = APPROVED)	Uploads attributed to an unverified company are worthless in an audit. Verification needs your proof of affiliation — institutional authorisation letter / GST / CIN, PDF or JPG/PNG, ≤5 MB, hashed at upload, never reprocessed
A named person with authority to commit (designation recorded: General Manager (CSR), Head – CSR (Jharkhand), CEO)	FUNDER credit edges and MoU drafts cite a person, not a logo
District focus and domain interests on your org record	This is what makes a live Jharia EDUCATION challenge appear in your discover feed

1. Upload by stage

Four columns rather than seven, because a stage table that needs a landscape page is a stage table nobody reads. Each row names the artifacts.kind the file lands as, and the licence, in the second column.

# · stage	What you file (kind · licence)	Format	What the platform does with it
1 · Expression of interest — /industry/challenges/[trackingId]	No file. 20–3,000 characters saying what you will fund, the amount band, your constraint ("only within 25 km of a plant township", "no work with minors directly"), and who signs	text	industryInterests row at EXPRESSED; notifies the project lead. Silently rejected if the challenge has not cleared triage
2 · Accepted	Nothing yet	—	FUNDER credit edge on the project; challenge → INDUSTRY_INTEREST
3 · Agreement	Counter-signed MILAN MoU. The platform generates and hashes the draft; you upload the executed scan with signature and stamp	AGREEMENT · RESTRICTED · PDF ≤10 MB	PROPOSAL ledger entry carrying the file's own SHA-256; every read of it is logged
4 · Money	Sanction order or CSR-1 project code, committed ₹ amount, head of account, disbursement schedule, grant vs in-kind (hardware at invoice price)	FUNDING · RESTRICTED · PDF ≤10 MB	PROPOSAL ledger entry. Amounts stay claims until the pilot closes; the CSR export shows them beside confirmed outcomes, never summed into one total

# · stage	What you file (kind · licence)	Format	What the platform does with it
5 · Technical basis	BOM with unit prices, vendor + warranty terms, the datasheets you are buying against, earthing/lightning/IS-compliance notes, and what you are explicitly not fixing ("no new poles; existing 46 only")	SPEC · CC_BY · PDF ≤25 MB	REPORT ledger entry. Public by default — a funder's spec is checkable by the university that has to satisfy it
6 · As-built	Per-pole record: asset tag, GPS of the pole (never a home), commissioning date, energisation readings (voltage, lumens or load current), before/after photos with faces and name-boards blurred, contractor sign-off	REPORT · CC_BY · PDF/JPG ≤10 MB each	REPORT per file, each keyed by content hash, so re-uploading the same bytes deduplicates instead of double-counting
7 · Test & acceptance	30-day and 90-day uptime logs from the retrofit telemetry, failure count, and the acceptance certificate signed by the school and the Panchayat — they are the owner; if they have not accepted it, it is not done	REPORT · CC_BY · PDF	REPORT. This is the file that turns "installed" into "verified by a counterparty"
8 · Sustainability	The O&M instrument: AMC order or spares-corpus receipt with ₹/pole/year, the named electrician and phone number, spares stock list, trained-repair register, escalation path when the money runs out	REPORT · CC_BY · PDF	REPORT. File it at sanction, not at close — a plan written after the fact is a footnote
9 · Capacity	Training record for the repair crew: date, attendance count (aggregate, no names), materials issued, refresher schedule	REPORT · CC_BY · PDF	REPORT
10 · Impact claim	markImplemented() — a form, not a file: one sentence on what changed, plus pointers to the evidence above	text	IMPLEMENTED; notifies every corroborator and the reporter with a signed /me/verify/[token] link needing no login, and opens IMPACT_UNCONFIRMED_30 so silence is re-asked, never assumed
11 · Close-out	Third-party or internal audit note; the year's CSR-1 extract; a corrected figure if the community disputed anything	REPORT · RESTRICTED if it carries financial detail · PDF	REPORT. Auditors are the intended reader — say so in the abstract field

2. What the platform refuses (and returns with a reason)

- **Images of identifiable children.** Any photograph where a face is recognisable, in uniform, at a school, or at a work site. Faces must be blurred by the uploader before upload. Under the DPDP Act 2023 these are personal data of minors processed for a purpose that does not require them, and a CSR report is not a consent mechanism.
- **Named beneficiary lists.** Attendance rosters, subsidy beneficiary tables, family names, mobile numbers, Aadhaar, bank details, ration-card images. Upload **counts and aggregates**; keep the roster in your own audited system and cite it.
- **Home locations.** GPS is welcome for assets (the pole, the pump, the classroom). Never for households — in a coalfield resettlement context that is a mapping of who lives where, and it will be returned.
- **Anything you don't have the right to publish.** Vendor datasheets and bid documents are typically the funder's to share; a contractor's drawing set often is not. Upload what you own or are licensed to distribute, and choose RESTRICTED when in doubt — a restricted artifact is still readable

by a named person who states a purpose, and every read is logged.

- **Unverifiable outcome language.** "Transformed the lives of 500 children" in a file titled evidence fails review; the same sentence belongs in your annual report, not the ledger.
- **Marketing collateral, logos-only PDFs, press clippings.** Not an artifact type.
- **Duplicate bytes under a new title.** Same content hash → same object. Renaming does not create two results, and the ledger entry will not say that it did.

3. File discipline (six rules that cost you nothing)

1. **Name:** JH-2026-DHN-####_<stage>_<asset|site>_YYYY-MM-DD.pdf — the tracking ID first, so a file is attributable even after it leaves the platform.
2. **Title field:** plain and specific — "46-pole lighting retrofit, BOM and unit prices, DVC-CSR/2026/114" beats "Project report".
3. **Abstract field:** three sentences — what it proves, over what period, and what it explicitly does **not** cover. Researchers and auditors read this line and nothing else.
4. **Checksum:** let the platform compute it. If you maintain an internal register, record the returned SHA-256 — it is what proves your copy and the public copy are the same file.
5. **Dates, not "recent".** Every claim needs an as-of date; the SLA ladder and the annual review both read dates.
6. **One claim per file.** A PDF containing the BOM, the invoices and the impact story is three ledger entries pretending to be one, and every future correction to the invoices means re-publishing the impact story too.

4. A good EOI (copy the shape, not the words)

DVC CSR wing can fund the lighting component for the four Jharia night-school centres: retrofit controllers, batteries for the 20 poles with degraded storage, and a 2-year spares corpus, in the range ₹12-18 lakh, subject to the block Panchayat accepting O&M ownership in writing. We can start disbursal within 6 weeks of a signed MoU. Our constraint: works only on existing poles — no new civil works, no land or ROW questions. Our signatory: GM (CSR), DVC. We would like the university team to carry out the 30/90-day uptime measurement so that the acceptance data is not self-generated.

Twenty seconds to read; four checkable commitments; one honest constraint; and a request that the evaluator not be the funder. That last clause is why a project lead accepts this one and declines the glossy one.

5. What you get back

- **A FUNDER credit edge on a citable project** — permanently attached to the artifact set, with a public /credit trail. It survives staff turnover, which is the entire reason an annual PDF is not an alternative.
- **A \$135 export an auditor can defend:** confirmed / partly-confirmed / claimed-but-not-confirmed as three separate counters, with beneficiary totals never summed. Your spend is not discounted by the honesty — it is the only part of the report that survives scrutiny.
- **Defensive publication of the university's design**, hashed: it does not stop someone filing a patent, it makes the work prior art, which is what stops one being granted over it.
- **A verification trail for the asset you funded** — every state change, every access to a restricted file, and every human override in one hash chain, recomputable in a browser.
- **Repeatable targeting:** your domain interests and district focus next year are matched against scored, corroborated, citizen-verified demand — not against whoever asks loudest.

6. One-line summary for the CSR head who asks why this is worth the effort

"You upload six PDFs you were going to produce anyway; what you get is that nobody — not a future department, not a future government, not your own successor — can quietly make the work, or your contribution to it, un-have happened."